

Dataset

Key concept: `case_id` (global primary key)

Across **all** provided files, the most important identifier is `case_id` :

- In `train.json` / `val.json` / `test.json` , the **top-level JSON key** (e.g. `"2146"`) is the `case_id` .
- In CSV files under `radiomics_info/` and `report_info/` , there is a column named `case_id` .

When joining different modalities/features/reports, always join on `case_id` .

Files

`train.json` / `val.json` / `test.json`

Each file is a JSON **dictionary**:

- **key**: `case_id` (string, e.g. `"2146"`)
- **value**: a record describing this case

Common fields in each record:

- **modality** : list of modality names for this case (e.g. `["ax_t2f", "ax_t2", "ax_t1c+", "ax_t1"]`)
- **available_modalities** : list of modalities actually available
- **image_path** : list of relative paths to `.npy` feature files (aligned with modalities)
- **report** : radiology report text (string)
- **Sex** , **Age** : basic demographics (strings)

Label field:

- **Overall_class** : **only present in `train.json` and `val.json`** (this is the classification label)
- In `test.json` **there is no `Overall_class`** (no labels provided to students)

Minimal example (from `train.json`):

```
{
  "2146": {
    "modality": ["ax t1", "ax t1c+", "ax t2"],
    "Overall_class": "Meningioma",
    "available_modalities": ["ax t1", "ax t1c+", "ax t2"],
    "image_path": [
      "image_features_filtered/2146/ax_t1/image.npy",
      "image_features_filtered/2146/ax_t1c/image.npy",
      "image_features_filtered/2146/ax_t2/image.npy"
    ],
    "report": "...",
    "Sex": "female",
    "Age": "51"
  }
}
```

sample_submission.csv

Kaggle submission template:

- **case_id** : test case id
- **Overall_class** : your predicted class for that **case_id**

Folders

image_features/

Per-case feature arrays saved as NumPy `.npy` .

Paths referenced by `image_path` are relative to this dataset folder.

radiomics_info/

Radiomics features in CSV format, organized by split:

- `radiomics_info/train/*.csv`
- `radiomics_info/val/*.csv`
- `radiomics_info/test/*.csv`

All radiomics CSVs contain a `case_id` column for joining.

report_info/

Structured patient/report info in CSV format, organized by split:

- `report_info/train_patient_info.csv`
- `report_info/val_patient_info.csv`
- `report_info/test_patient_info.csv`

All report CSVs contain a `case_id` column for joining.

Recommended workflow

1. Load `train.json` (or `val.json`) as the base table.
2. For each `case_id`, use `image_path` to load the `.npy` features.
3. Optionally join additional features from:
 - `radiomics_info/<split>/*.csv`
 - `report_info/<split>_patient_info.csv`
4. Train on `train.json`, validate on `val.json`, and predict `Overall_class` for `test.json`.